# PAPER_692: M51 Whirlpool Galaxy: Tidal Arm Formation and UQFF Star Formation

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## Abstract

The Whirlpool Galaxy (M51a / NGC 5194) and its companion M51b (NGC 5195) form a spectacular interacting pair 23 Mly away. Tidal forces from M51b drive the prominent spiral arm structure and enhance star formation rates.

## 1. System Parameters

| Parameter | Value |
|---|---|
| $M_{M51a}$ | $1.6 \times 10^{11} M_\odot$ |
| $M_{M51b}$ | $2 \times 10^{10} M_\odot$ |
| Separation | 7.5 kpc |
| SFR | 3 $M_\odot$/yr |

## 2. Tidal Force

$$F_{tidal} = \frac{2GM_{comp}R_d}{d_{sep}^3}$$

This differential force stretches M51a's disk and drives arm formation.

## 3. Spiral Arm Pitch Angle

$$i_{pitch} = \arctan\left(\frac{F_{tidal}}{v_c^2/R}\right)$$

## 4. UQFF MUGE for M51

$$g_{M51} = \frac{G(M_a + M_b)}{r^2}(1 + H(z)) + \rho_{ISM} V g \frac{\rho_{SCm}}{\rho_{UA}}$$

## 5. UQFF Star Formation Efficiency

$$\text{SFE}_{UQFF} = \text{SFR}\left(1 + \frac{F_{tidal}}{g_{grav}}\right)(1 + f_{TRZ})$$

## References

- Salo & Laurikainen (2000), MNRAS, 319, 393
- UQFF Framework, Star-Magic Session 174

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